Automatic positioning system of medical service robot based on binocular vision

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***Abstract*—Aiming at the problems of large difference and inaccurate positioning in the practical application of traditional positioning system, an automatic positioning system of medical service robot is designed based on binocular vision. Through the hardware structure design of camera, video codec device, host computer display, and the software structure design based on binocular vision to detect the coordinates of extreme points in space, match and locate the key direction of visual information in space, a new automatic positioning system is proposed. Compared with the traditional positioning system, the new positioning system has smaller positioning difference and higher positioning accuracy.**

Keywords-Binocular vision; Medical service robot; Automatic positioning system; Hardware design; Software design;

I. INTRODUCTION

In the in-depth market survey of the medical industry, it is found that low-cost and high-efficiency auxiliary medical services for the elderly living alone, the disabled and other special groups are the focus of research and attention in today's society, and are also the core issues of modern medical industry in the development and market construction [1]. Although the development and appearance of medical service robots in recent years have largely solved the problem of auxiliary services in the medical industry, most of the robots supplied to the market lack an intelligent system to support their operation. In the process of searching the relevant literature, it is found that many domestic and foreign technical researchers have made remarkable achievements in the design of intelligent positioning system for medical service robot [2]. But most of the algorithms used in the system are used in the specified space, which can not achieve the dynamic control of the system. Therefore, this paper puts forward the concept of binocular vision in the design of this aspect, which integrates the target of robot recognition, the work to be performed and the terminal positioning, and takes the human vision as the reference to locate the terminal information.

II. HARDWARE DESIGN

In this paper, the positioning system hardware design, the actual situation of the medical service robot in the working state, using the camera, video codec device, PC display as the main structure of the system hardware. The image and video of the position and working state of the robot in the medical service area are acquired through the camera. Therefore, the day and night HD camera ds-16518 (c) is selected, and the image resolution of the camera is 1280 × 1080, 1 / 3.6 "CMOS sensor, 5.5-85mm20x optical zoom lens [3]. The image captured by the camera is encoded by the video encoder, and the encoded video image is transmitted to the upper computer of the system in the form of digital signal by wireless transmission technology, and then the digital signal is converted into image video by the video decoder to complete the next operation. The video codec in this paper selects isde165-659 codec with HDMI, DVI and VGA three different formats, which has the advantage of H.264 professional HD and supports a variety of different HD signals. The network output interface of the codec is used to transmit the encoded signal [4]. At the same time, the system introduces an automatic control device on the basis of the traditional positioning system, so that it can accurately locate the position of the medical service robot in the first time when it is recognized that there is a mobile medical service robot in the working range of the system. At the same time, in order to ensure that in the follow-up system operation process, we can retrieve the image basis for the existing problems, and ensure that the video data obtained by the camera can be stored for a long time, this paper introduces the resource buffer in the system, which is used to realize the short-time storage of image and video information, and provide the basis for the follow-up positioning.

III. SOFTWARE DESIGN

A. Detection of spatial extreme point coordinates based on binocular vision

In order to ensure that the medical service robot can accurately identify the key information in the effective scale space, this chapter introduces the binocular vision theory, and generates the convolution image of equal scale size with reference to the Gaussian spatial distribution algorithm. In this process, the binocular vision detection range can be defined as $I(x,y)$, so it can be considered that the vision range of the medical service robot is the same as it [5]. At this time, through the simulation of binocular vision, a multi-scale image with spatial visual information can be obtained. On this basis, the convolution image is scaled, and the visual convolution kernel is used as the only channel to describe the feature parameters of the visual image. In the description process, it can be considered that the less evaluation information in convolution image, the lower its relative scale, and the clearer the corresponding visual image details. Based on the above analysis, the recognizable images of the medical service robot in the effective visual range are described. The calculation formula is as follows.

$$\begin{aligned} D(x,y,\sigma) &= \left[G(x,y,k\sigma)-G(x,y,\sigma)\right]\times I(x,y) \\ &= L(x,y,k\sigma)-L(x,y,\sigma) \end{aligned} \quad (1)$$

In formula (1): x is the abscissa of binocular vision sampling point; y is the ordinate of the sampling point of binocular vision; σ is the Gauss difference parameter; k is the scale of visual image; D represents the image that can be recognized by the medical service robot in the effective visual range; G represents visual images of different scale sizes; I is the general image; L is the adjacent image in two-dimensional space.

After completing the calculation of formula (1), it detects whether the coordinate point is at the edge position of the generated image. In the process of detection, firstly, it should be clear whether the output coordinate point is the point coordinate with relatively low contrast in the visual image; Secondly, it should be clear whether the point coordinates have high response ability to the image edge information; Finally, the point coordinates with features are extracted by dog operator [6]. And through multiple 2D and 3D fitting of point coordinates, the response ability of extreme points is increased. The extraction process of extreme points is shown in Figure 1.

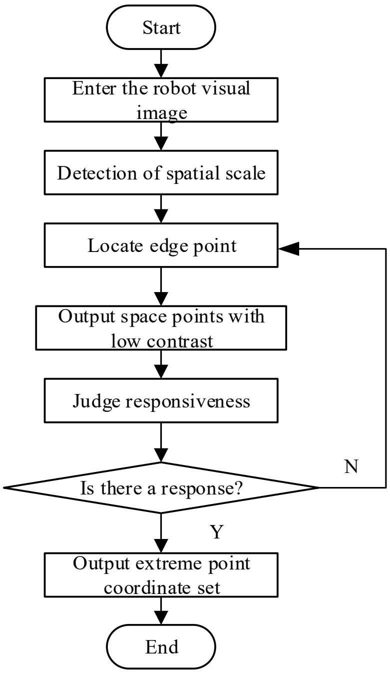

Figure 1. Extremum extraction process of visual image

According to the process described in Figure 1 above, the coordinate set of extreme points in binocular vision detection space is output.

B. Matching and locating key directions of visual information in space

In order to enhance the accuracy of the automatic positioning of the medical service robot, it is necessary to match the key directions of the visual information in the space after the extraction of the extreme points in the visual image. In the process of image matching, 4.0 is used for each key data identified × 4.0. At this time, 128.0 data can be generated for each key point in the visual range, and each data represents a dimension feature [7]. In order to ensure the accuracy of dimensional features describing transactions, we need to generate scale invariant feature transformation operators to ensure the normalization of positioning results. The description of eigenvectors is shown in Figure 2.

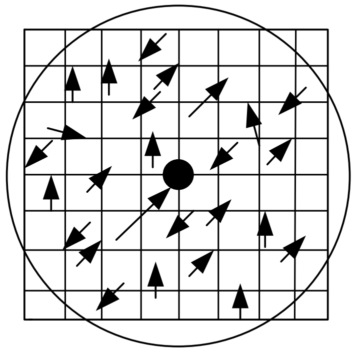

Figure 2. Feature vector describing visual information

The key information in Figure 2 is extracted, and the key vectors are matched by size adjustment, spatial rotation, geometric transformation and other methods. The matching effect is shown in Figure 3.

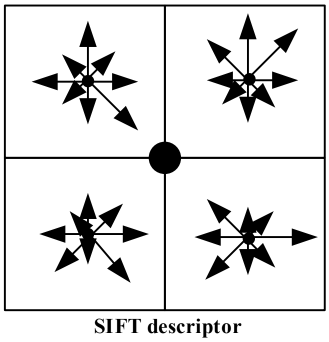

Figure 3. Key directions of visual information in matching space

According to the contents of Figure 2 and figure 3 above, the key directions of visual information in space are matched. On this basis, the spatial positioning coordinates of the key direction in Figure 2 are output. Combined with the effective direction of the coordinate extremum in space, the specific coordinates are output to complete the positioning of the key direction of visual information and realize the operation of the automatic positioning system function of medical service robot.

IV. CONTRAST EXPERIMENT

A. Experimental preparation

Combined with the above discussion, the hardware and software design of the automatic positioning system of medical service robot based on binocular vision is completed. In order to verify whether the system can effectively solve the problems existing in the practical application of traditional positioning system, the following comparative experiments are carried out

In order to ensure the objectivity of the comparative experiment, the target tracking system based on EKF algorithm in the service robot in reference [2] and the system designed in this paper are run in the same experimental environment, and the same type of medical server robot is selected. Table 1 is the comparison table of the technical parameters of the medical server robot in this experiment.

TABLE I. COMPARISON TABLE OF TECHNICAL PARAMETERS OF EXPERIMENTAL MEDICAL SERVER ROBOT

Serial number	Skill parameters	index
1	structure size	188mm×125mm
2	weight	2.5kg
3	running speed	55.8cm/s
4	Battery	6.4VRated voltage, can be recharged 500 times
5	Continuous working hours	More than 1.5h
6	Series Rate	7200bps

B. *Positioning error test*

In the process of the experiment, the medical service robot completes its operation, and uses two kinds of positioning systems to locate the robot. The positioning cycle of the two systems is set to be 1H / time. The positioning results of the two systems are compared with the coordinates of the actual position of the robot, and the positioning difference is calculated. The actual positioning experiment is carried out by using the system. In the process of positioning experiment, the accuracy of the medical server robot positioning system is mainly judged by the coordinate error of the target person. The coordinate data of medical server robot positioning system is shown in Table 2.

TABLE II. COORDINATE DATA

Serial number	Actual positioning coordinates	Coordinate data	
		design system	Literature [2] System
01	(-0.562, -0.285, 0.929)	(-0.562, -0.285, 0.929)	/
02	(-12.759, -21.582, 16.378)	(-12.759, -21.582, 16.378)	(-12.759, -21.582, 16.378)
03	(8.593, -5.248, 8.516)	(8.593, -5.248, 8.516)	(8.593, -5.248, 8.516)
04	(15.584, 8.571, 0.719)	(15.584, 8.571, 0.719)	(15.584, 8.571, 0.719)
05	(-6.432, -1.371, 23.083)	(-6.419, -1.370, 23.083)	/

Based on the data in Table 1, the error between the actual positioning results and the system positioning results in reference [2] is further analyzed. The positioning error is the distance difference between the measured coordinates and the actual coordinates. The specific experimental results are shown in Table 3

TABLE III. POSITIONING ERROR

Serial number	positioning error/ (mm)	
	design system	Literature system
01	0.05	2.87
02	0.10	4.69
03	0.15	5.29
04	0.20	6.32
05	0.32	7.38

According to the analysis of the experimental data in Table 3, the maximum positioning error of the design system is 0.32mm, which is lower than the comparison method. The minimum positioning error of the system in reference [2] is 2.87mm, which is higher than the maximum value of 0.32 of the design system. This is because the design system adds binocular vision method in the process of target recognition and positioning, deeply analyzes the coordinates of the target image, obtains the key information of the target image, effectively reduces the benchmark displacement in the test, and is closer to the real results.

In order to compare the results of the two positioning systems more clearly, this paper makes statistics of the positioning difference between the two systems within 5 hours, and draws the experimental results comparison diagram as shown in Figure 4.

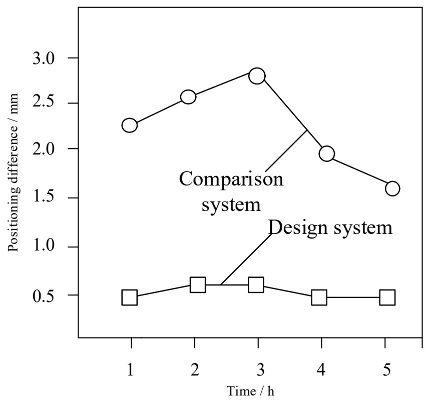

Figure 4. Comparison of experimental results of two positioning systems

The design system in Figure 4 is the automatic positioning system of medical service robot based on binocular vision proposed in this paper, and the comparison system is the positioning system in reference [2]. It can be seen from the experimental results in Fig. 4 that the positioning error of the design system is obviously smaller than that of the comparison system in five times of positioning. Therefore, through comparative experiments, it is proved that the automatic positioning system of medical service robot based on binocular vision proposed in this paper has higher positioning accuracy in practical application, and can accurately locate the medical service robot.

C. *Time cost test*

Time cost refers to the time consumed by medical service robot localization. In order to accurately mark the time cost data, the medical service robot localization is divided into two stages (experimental array 01 ~ 05), which are feature extraction stage (experimental array 01 ~ 05) and localization stage (experimental array 06 ~ 10). The total time consumed in the two stages is the time cost. The independent variable is the number of experiments and the dependent variable is the time cost.

According to the analysis of the experimental data in Table 4, the positioning time range of the medical service robot in reference [2] is 19.56s-29.45s; The positioning time range of medical service robot is 9.08s-15.46s. Therefore, the positioning time of the medical service robot in the design system is less than that of the comparison method, which proves that the positioning effect of the design system is better, and the application performance of the binocular vision method is better.

V. CONCLUSION

Compared with the conventional algorithm used in the system, the algorithm proposed in this paper does not need to predict the trajectory of the dynamic target in the visual range, and does not need to master the transaction characteristics of the target, which can effectively realize the recognition and positioning of the dynamic target trajectory. However, the related research is still in the stage of development in the medical market, and there are no results to prove that the proposed theory has a certain basis. Therefore, this paper will combine the binocular vision theory, from the two aspects of hardware and software, to design and research the automatic positioning system of medical service robot. And after the completion of the design of this system, through comparative experiments, it is proved that the design results have significant advantages, which can meet the needs of the medical industry.

In the process of the experiment, the medical service robot completes its operation, and uses two kinds of positioning systems to locate the robot. The positioning cycle of the two systems is set to be 1H / time. The positioning results of the two systems are compared with the coordinates of the actual position of the robot, and the positioning difference is calculated. In order to compare the results of the two positioning systems more clearly, this paper makes statistics of the positioning difference between the two systems within 5 hours, and draws the experimental results comparison diagram as shown in Figure 4.

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